LECTURE 2

Sampling

How to sample effectively, and how to quantify the samples we collect.

Presented by Dr. Devan Becker, Data200

Adapted from Data 100, Summer 2023 @ UC Berkeley by Bella Crouch and Dominic Liu

Content credit: Acknowledgments

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Announcements

- Friday is the last day to drop with absolutely no repercussions.

- Monday's participation is due tonight!
  - Today will use iClicker - not too late to register!
  - Note: iClicker subscriptions are more expensive in the Apple app.

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Today's Roadmap

- Censuses and Surveys
- Sampling: Definitions
- Sampling Bias: A Case Study
- Probability Samples
- Multinomial Probabilities

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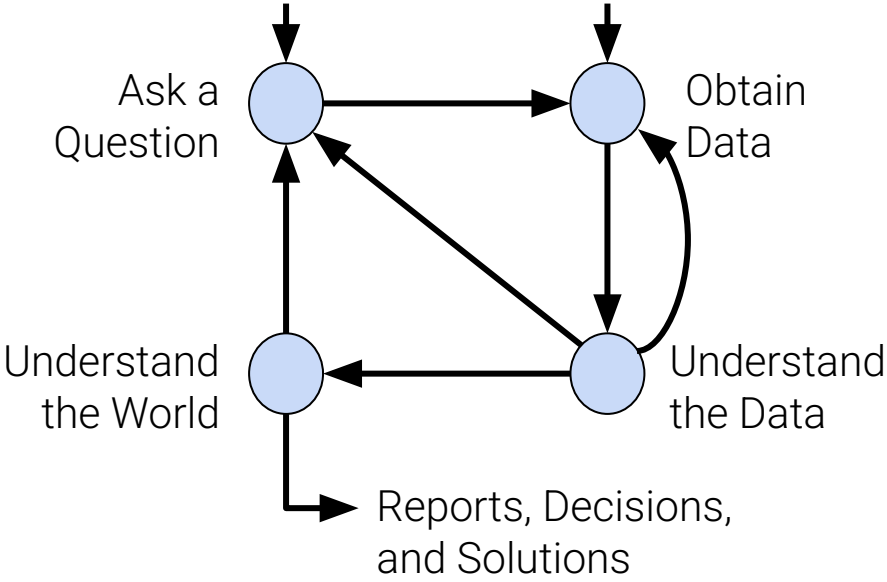
Recall the
Data Science Lifecycle.



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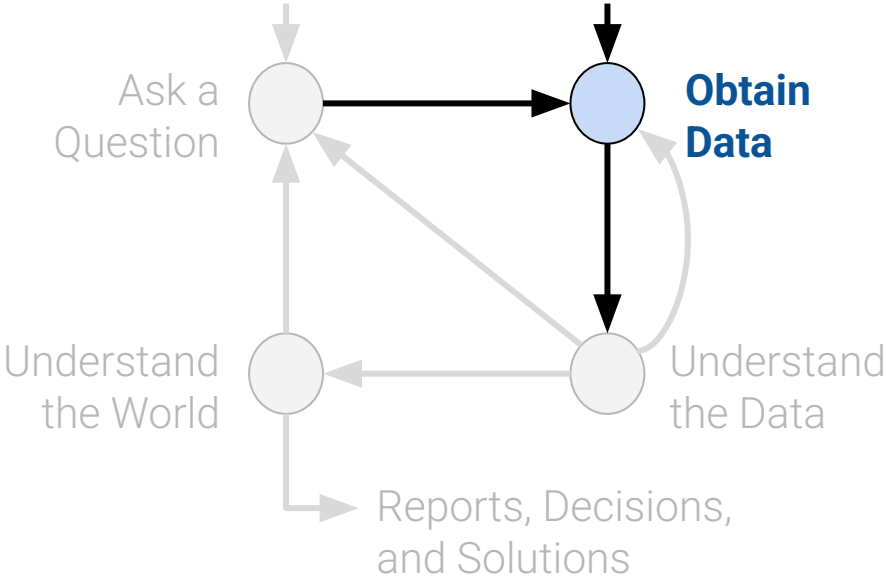
Today

How do we collect data?



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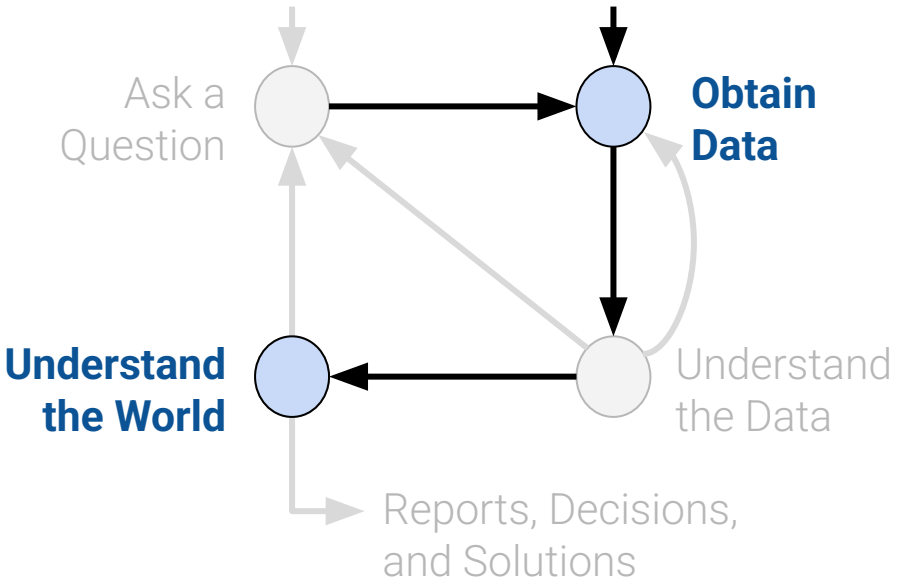
Today

How does understanding data collection help us understand the world?



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Censuses and Surveys

- **Censuses and Surveys**
- Sampling: Definitions
- Sampling Bias: A Case Study
- Probability Samples
- Multinomial Probabilities

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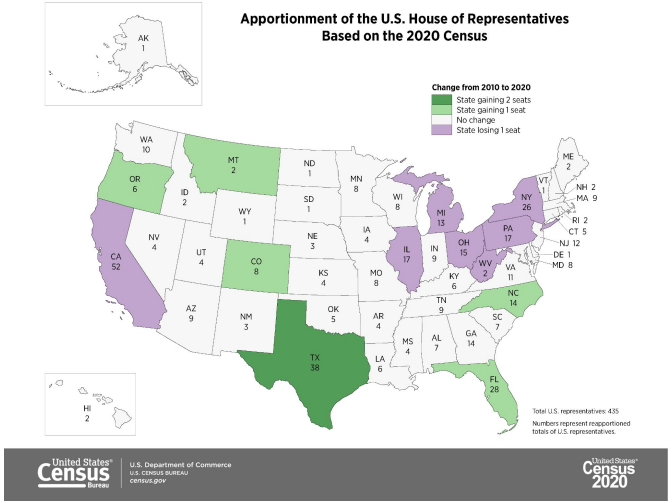
Census

A **census** is “an official count or survey of a **population**, typically recording various details of individuals.”

The US Decennial Census

- Was last held in April 2020.
- Counts **every person** living in all 50 states, DC, and US territories. (Not just citizens.)
- Mandated by the Constitution. Participation is required by law.
- Important uses:
  - Allocation of Federal funds.
  - Congressional representation.
  - Drawing congressional and state legislative districts.



data.census.gov

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Survey

Recall: a **census** is "an official count or **survey** of a population, typically recording various details of individuals."

A **survey** is a set of questions.

- For instance: census workers survey individuals and households.

What is asked, and how it is asked, can affect:

- How the respondent answers.
- **Whether** the respondent answers.

There are entire courses on surveying!

FiveThirtyEight

Politics Sports Science Podcasts Video Interactives

JUN. 27, 2019, AT 12:42 PM

The Supreme Court Stopped The Census Citizenship Question — For Now

NATIONAL

Citizenship Question To Be Removed From 2020 Census In U.S. Territories

August 9, 2019 · 3:23 PM ET

FiveThirtyEight, NPR

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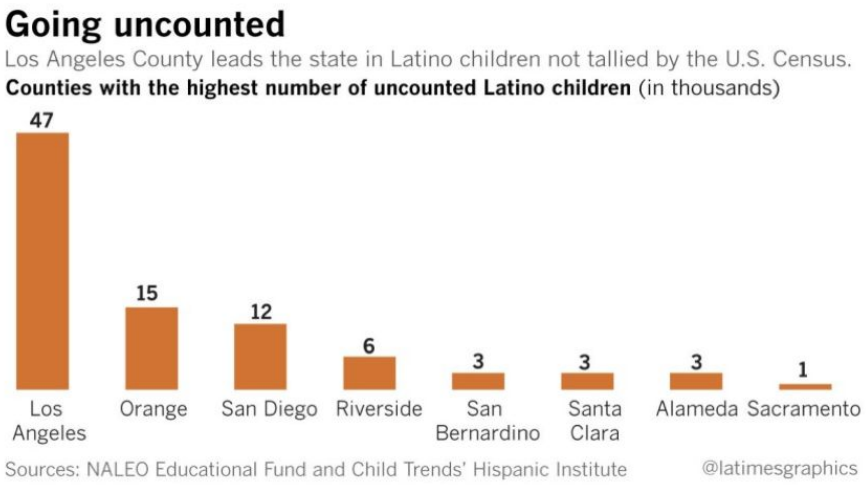
Undercounting in the US Decennial Census

LA Times 2010 Census



How do we know these numbers?
Other surveys

WaPo 2000 Census

High Court Rejects Sampling In Census
Ruling Has Political, Economic Impacts

Sampling methods would estimate Americans who missed the survey.
— Most often minorities/poor who vote Dem.
— "The better way is to improve the methods for contacting and questioning every household"

NY Times 2020 Census

In 2020 Census, Big Efforts in Some States. In Others, Not So Much.

California is spending $187 million to try to ensure an accurate count of its population. The Texas Legislature decided not to devote any money to the job. Why?

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Sampling: Definitions

- Censuses and Surveys
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Sampling from a finite population

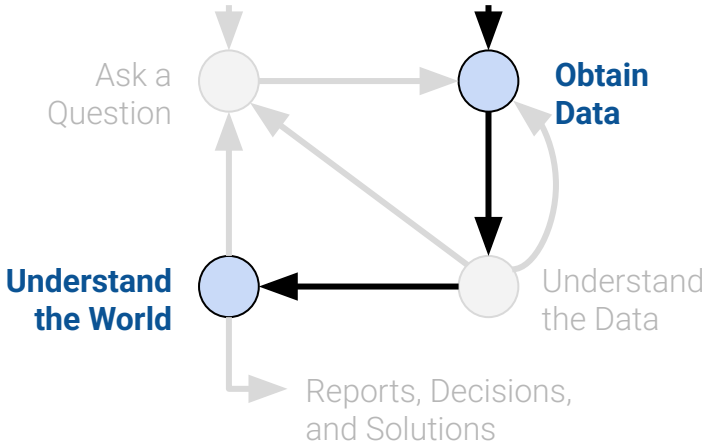
A census is great, but expensive and difficult to execute.

- Would **all** voters be willing to participate in a voting census prior to an actual election?

A **sample** is (usually) a subset of the population.

- Samples are often used to make **inferences about the population**.
- How you draw the sample will affect your accuracy.
- Two common sources of error:
  - **chance error**: random samples can vary from what is expected, in any direction.
  - **bias**: a systematic error in one direction.
    - Could come from our sampling scheme, and survey methods.

Inference: quantifying degree of certainty in our models of the world.
Data 8 book



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Population, sample, and sampling frame

Population: The group that you want to learn something about.

Sampling Frame: The list from which the sample is drawn.

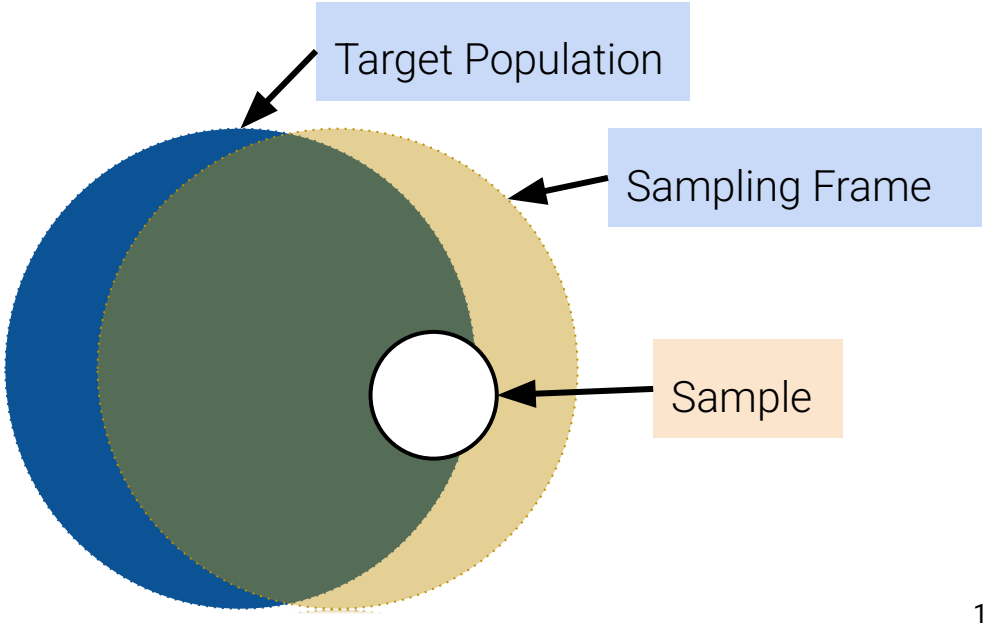
- If you're sampling people, the sampling frame is the set of all people that could possibly end up in your sample.

Sample: Who you actually end up sampling.

- A subset of your sampling frame.

There may be individuals in your **sampling frame** (and hence, your sample) that are **not** in your population!

Similarly, there might be individuals in your target population that are not in your sampling frame.



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Other kinds of populations

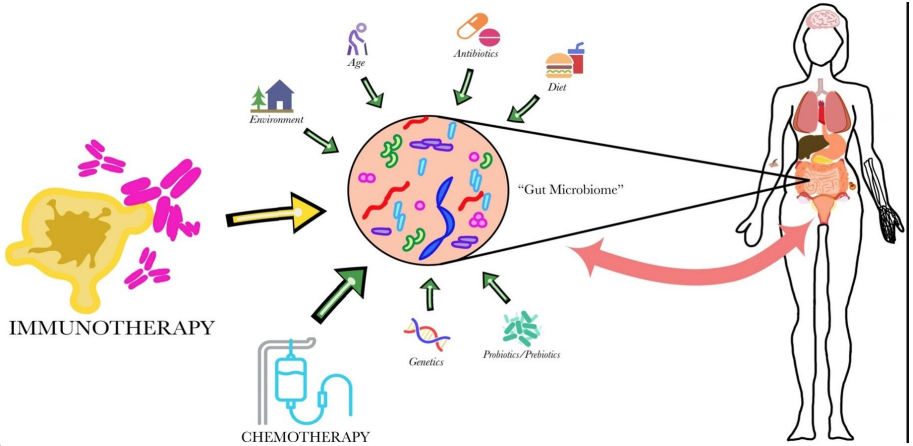
The individuals in a population are not always people!

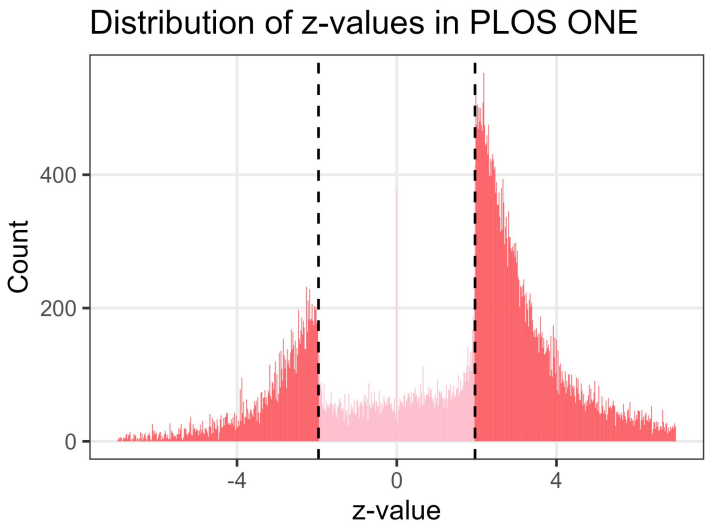
Could be

- **Bacteria** in your gut (sampled using DNA sequencing)
- **Trees** of a certain species
- **Small businesses** receiving a microloan
- **Published results** in a journal / field (example)

In any of these cases we might examine a sample and try to draw an inference about the population it came from.

- Simplest example: what % have some binary property (like voting intention)?





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Bias: A Case Study

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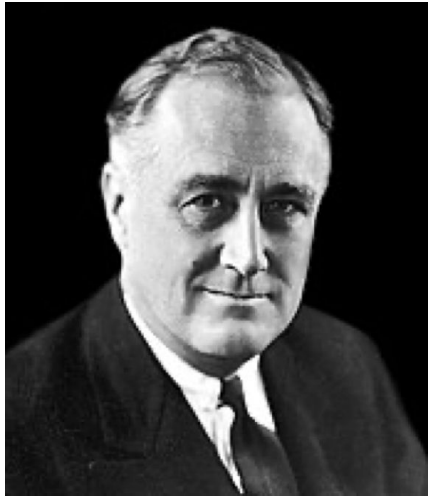
Case study: 1936 Presidential Election



Roosevelt (D)

Landon (R)

In 1936, President Franklin D. Roosevelt (left) went up for re-election against Alf Landon (right). As is usual, **polls** were conducted in the months leading up to the election to try and predict the outcome.

(Election result spoiler: Landon was not a U.S. President)

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The Literary Digest: Election Prediction

The *Literary Digest* was a magazine. They had successfully predicted the outcome of 5 general elections coming into 1936.

They sent out their survey to **10,000,000** individuals, who they found from:

- Phone books.
- Lists of magazine subscribers.
- Lists of country club members.

	% Roosevelt	# surveyed
Actual election	61%	**All voters** (~45,000,000)
The Literary Digest poll	43%	10,000,000

How could this have happened?
They surveyed 10 million people!

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The Literary Digest: What happened?

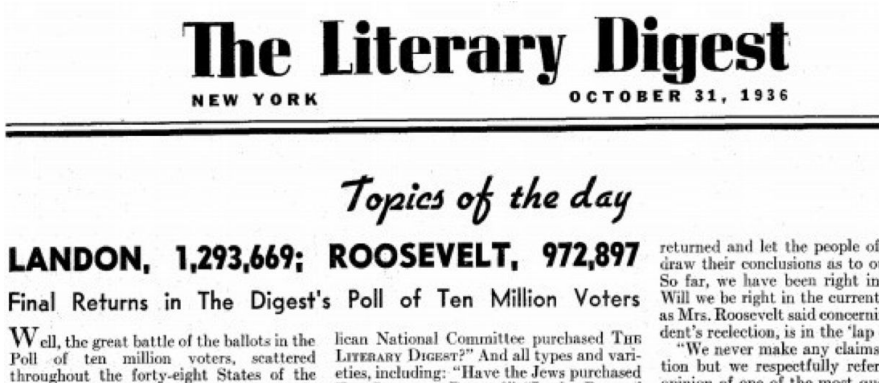
(1) The Literary Digest sample was **not representative** of the population.

- The Digest's **sampling frame**: people in the phonebook, subscribed to magazines, and went to country clubs.
- These people were more affluent and tended to vote Republican (Landon).

	% Roosevelt	# surveyed
Actual election	61%	**All voters** (~45,000,000)
The Literary Digest poll	43%	10,000,000

(2) Only 2.4 million people **actually filled out the survey!**

- 24% response rate (low).
- Who knows how the 76% **non-respondents** would have polled?



The Literary Digest

NEW YORK OCTOBER 31, 1936

Topics of the day

LANDON, 1,293,669; ROOSEVELT, 972,897

Final Returns in The Digest's Poll of Ten Million Voters

Well, the great battle of the ballots in the Poll of ten million voters, scattered throughout the forty-eight States of the

lican National Committee purchased THE LITERARY DIGEST?" And all types and varieties, including: "Have the Jews purchased

returned and let the people of
draw their conclusions as to o
So far, we have been right in
Will we be right in the current
as Mrs. Roosevelt said concerni
dent's reelection, is in the 'lap
"We never make any claims
tion but we respectfully refer

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Gallup's Poll: Election Prediction

George Gallup, a rising statistician, also made predictions about the 1936 elections.

His estimate was **much** closer despite having a smaller **sample size** of "only" 50,000

(Also more than necessary!)

George Gallup also predicted what The Literary Digest was going to predict, within 1%, with a **sample size of only 3000 people**.

- He predicted the Literary Digest's **sampling frame** (phonebook, magazine subscribers, country clubs).
- So he sampled those same individuals!

	% Roosevelt	**# surveyed**
Actual election	**61%**	**All voters** (~45,000,000)
The Literary Digest poll	43%	10,000,000
George Gallup's poll	56%	50,000
George Gallup's prediction of Digest's prediction	44%	3,000

Samples, while convenient, are subject to chance error and **bias**.

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Adapted from *Statistics* by Freedman, Pisani, and Purves, W.W. Norton NY

Common Biases

Selection Bias

- Systematically excluding (or favoring) particular groups.
- **Example**: The Literary Digest poll excludes people not in phone books.
- **How to avoid**: Examine the sampling frame and the method of sampling.

Response Bias

- People don't always respond truthfully.
- **Example**: Asking citizenship questions on the census survey→illegal immigrants might not answer truthfully
- **How to avoid**: Examine the nature of questions and the method of surveying.

Non-response Bias

- People don't always respond → People who don't respond aren't like the people who do!
- **Example**: Only 2.4m out of 10m people responded to The Literary Digest poll.
- **How to avoid**: Keep your surveys short, and be persistent.

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The Gallup Poll today

Gallup U.S. Election polls:

- **Sampling Frame**: “civilian, non-institutionalized population” of adults in telephone households in continental US
- **Random Digit Dialing** to include both listed/unlisted phone numbers (to avoid **selection bias**)
- **Within household selection process** to randomly select if ≥1 adult in household
  - If no answer, recall multiple times (to avoid **non-response bias**)

Gallup Poll

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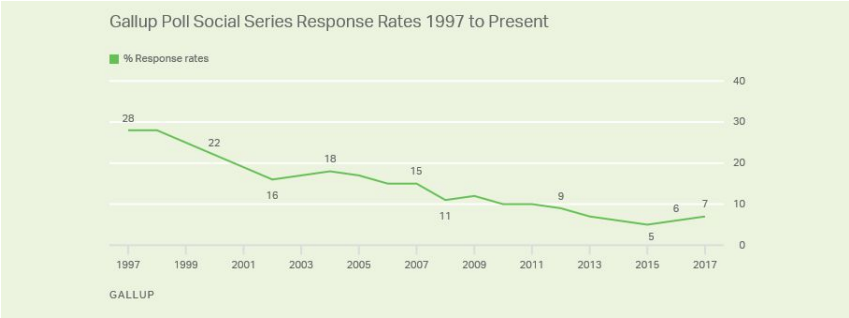
Election polling is hard!

Many sources of bias:

- **Who responds** to polls?
- Do voters **tell the truth?**
- How can we **predict turnout?**

Single-digit response rates are the norm



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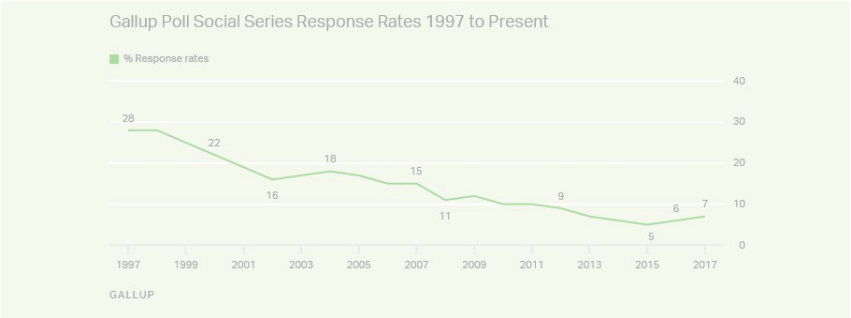

Poll numbers we see in the news are filtered through **proprietary statistical algorithms** that re-weight respondents
→ **"house effects"** of different pollsters

2022 ELECTION

Will The Polls Overestimate Democrats Again?

By Nate Silver
SEP. 16, 2022, AT 6:00 AM

FiveThirtyEight (source)

Polling bias isn't consistent

Weighted-average statistical bias in polls in final 21 days of the campaign

	PRES.	STATE LEVEL			
CYCLE	GENERAL	GOVERNOR	U.S. SENATE	U.S. HOUSE	COMBINED
1998	–	R+5.8	R+4.5	R+0.9	R+3.8
1999-2000	R+2.4	R+0.2	R+2.8	D+1.2	R+1.8
2001-02	–	D+3.5	D+2.0	D+1.4	D+2.6
2003-04	D+1.1	D+1.9	D+0.8	D+2.1	D+1.4
2005-06	–	D+0.4	R+2.1	D+1.1	D+0.1
2007-08	D+1.0	R+0.1	D+0.1	D+1.4	D+0.9
2009-10	–	R+0.2	R+0.8	D+1.3	D+0.4
2011-12	R+2.5	R+1.6	R+3.1	R+3.2	R+2.8
2013-14	–	D+2.3	D+2.7	D+3.9	D+2.8
2015-16	D+3.3	D+3.1	D+2.8	D+3.4	D+3.0
2017-18	–	R+0.9	EVEN	R+0.8	R+0.5
2019-20	**D+4.2**	**D+5.6**	**D+5.0**	**D+6.1**	**D+4.8**
All years	D+1.3	D+0.9	D+0.7	D+1.2	D+1.1

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Probability Samples

- Censuses and Surveys
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Quality, not quantity!

A **huge sample size** does not fix a **bad sampling method**!

We want the sample to be **representative** of the population.

Think about **a bad toaster**: if it's **burning parts of the bread and leaving others untoasted**, putting it in for longer won't result in an even toast!

- If your sampling is BAD, and your sample is BIG, what you'll have is a BIG BAD sample

Easiest way to to get a representative sample is by using **randomness**.

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Non-random sample: Convenience Samples

(source)

An example of a non-random sample is **convenience sample.** It's whatever we can get a hold of.

Example: Scientists in New South Wales (AUS) collect specimens from eucalyptus trees to keep in museums, recording **where they came from** in latitude / longitude.

*Can we use this data to map the **geographic distribution** of eucalyptus trees?*

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Convenience Samples

(source)

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Example: Scientists in New South Wales (AUS) collect specimens from eucalyptus trees to keep in museums, recording **where they came from** in latitude / longitude.

*Can we use this data to map the **geographic distribution** of eucalyptus trees?*

Warning:

- Haphazard ≠ **random**.
- Many potential sources of bias!

Like polls, we can try to correct bias by statistical modeling.
But it's better if we don't have to!

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Probability Sample (aka Random Sample)

Why sample at random?

1. (As mentioned before) To get more representative samples → **reduce bias**
   - Random samples **can** produce biased estimates of population quantities.
2. More importantly, with random samples we can **estimate** the **bias** and **chance error** → **quantify uncertainty**

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Probability Sample (aka Random Sample)

Why sample at random?

1. (As mentioned before) To get more representative samples → **reduce bias**
   - Random samples **can** produce biased estimates of population quantities.
2. More importantly, with random samples we can **estimate** the **bias** and **chance error** → **quantify uncertainty**

For a **probability sample**,

- We have to be able to provide the **chance** that any specified **set** of individuals will be in the sample.
- All individuals in the population **need not** have the same chance of being selected.
- Because we know all the probabilities, we will be able to **measure the errors**.

The real world is usually more complicated!

- Election polling: When Gallup calls, most people don't answer.
- Bacteria: We don't know the probability a given bacterium will get into a microbiome sample.

If the sampling / measurement process isn't fully under our control, we try to **model it**.

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Example Scheme 1: Probability Sample

Suppose I have 3 TA's (**A**lan, **B**ennett, **C**eline):
I decide to sample <u>2 of them</u> as follows:

- I choose **A** with probability 1.0
- I choose either **B** or **C**, each with probability 0.5.

All subsets of 2:	{**A**, **B**}	{**A**, **C**}	{**B**, **C**}
Probabilities:	0.5	0.5	0

This is a **probability sample** (though not a great one).

- Of the 3 people in the population, I know the chance of getting each subset.
- Suppose I'm measuring the average distance TA's live from campus.
  - This scheme does not see the entire population!
  - My estimate using the single sample I take has some **chance error** depending on if I see AB or AC.
  - This scheme **biases** towards A's response

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Common random sampling schemes

A **random sample with replacement** is a sample drawn **uniformly** at random **with** replacement.

- Random doesn't always mean "uniformly at random," but in this specific context, it does.
- Some individuals in the population might get picked more than once

A **simple random sample (SRS)** is a sample drawn **uniformly** at random **without** replacement.

- **Every individual (and subset of individuals) has the same chance of being selected.**
- Every pair has the same chance as every other pair.
- Every triple has the same chance as every other triple.
- And so on.

A raffle could use either sampling scheme, depending on if winners are eligible for multiple prizes.

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Example Scheme 2: Simple Random Sample?

Consider the following sampling scheme:

- A class roster has 1100 students listed alphabetically.
- Pick one of the first 10 students on the list at random (e.g. Student 8).
- To create your sample, take that student and every 10th student listed after that (e.g. Students 8, 18, 28, 38, etc).

1. Is this a probability sample?

2. Does each student have the same probability of being selected?

3. Is this a simple random sample?

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Example Scheme 2: Simple Random Sample?

Consider the following sampling scheme:

- A class roster has 1100 students listed alphabetically.
- Pick one of the first 10 students on the list at random (e.g. Student 8).
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1. Is this a probability sample?

Yes.

For a sample [n, n + 10, n + 20, …, n + 1090], where 1 <= n <= 10, the probability of that sample is 1/10.

Otherwise, the probability is 0.

Only 10 possible samples!

2. Does each student have the same probability of being selected?

Yes.

Each student is chosen with probability 1/10.

3. Is this a simple random sample?

No.

The chance of selecting (8, 18) is 1/10; the chance of selecting (8, 9) is 0.

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Summary

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Data from Surveys and Censuses

- Census is *not* always the best way to do things.
  - Might only need a subset of the population

- Surveys can be *excellent*
  - Great statistical properties, even with somewhat small sample sizes.

- **Biases** are hard to avoid!
  - Mismatch of sampling and target frames
  - Non-response for a reason
  - Sample might not represent population

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Evaluating the Data in General

- Are these the data we need?
  - Are these *all* the data we need?
- Are these the data we expected?
- Do these data represent the population we're interested in?
- Who collected it and why?
- ...

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Multinomial Probabilities

- Censuses and Surveys
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- Probability Samples
- **Multinomial Probabilities**

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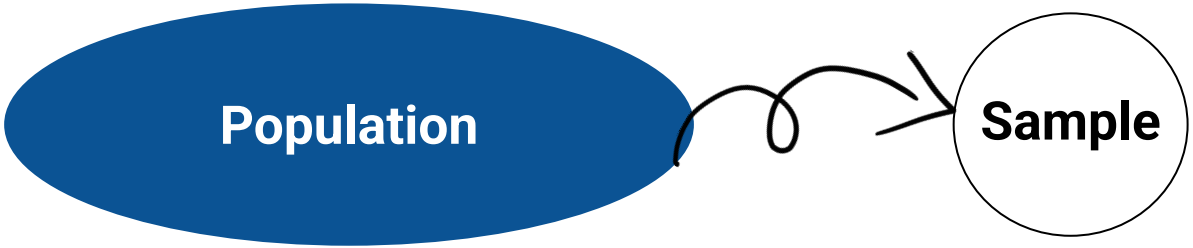
So far



If we have a probability sample:

- We can quantify error and bias.
- **Given the population distribution**, we can compute the probability of us getting a **particular sample**.

Note: We almost **never** know the population distribution! But this is a good start.

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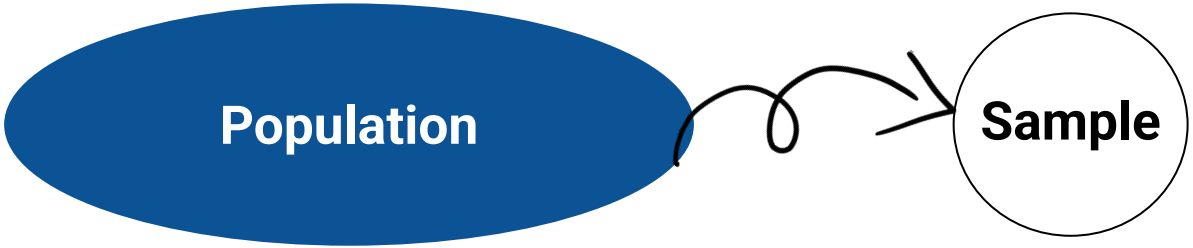
So far



If we have a probability sample:

- We can quantify error and bias (to be covered later).
- **Given the population distribution**, we can compute the probability of us getting a **particular sample**.

Note: We almost **never** know the population distribution! But this is a good start.

Special case: Random sampling with replacement of a **categorical population** produces **Multinomial Probabilities**.

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A very common approximation for sampling

A common situation in data science:

- We have an enormous population.
- We can only afford to sample a relatively small number of individuals.

If the **population is huge** compared to the sample, then
random sampling **with** and **without** replacement are pretty much the **same**.

Example: Suppose there are 10,000 people in a population.
Exactly 7,500 of them like Reese's.

What is the probability that in a random sample of 20, **all people like Reese's**?

SRS (Random Sample Without Replacement)

$$\left(\overbrace{\frac{7500}{10000}}^{0.75}\right)\left(\overbrace{\frac{7499}{9999}}^{0.74997}\right)\cdots\left(\overbrace{\frac{7482}{9982}}^{0.7495}\right)\left(\overbrace{\frac{7481}{9981}}^{0.7495}\right) \approx .003151$$

Random Sample With Replacement

$$\left(\frac{7500}{10000}\right)^{20} \approx .003171$$

Probabilities of sampling with replacement are much easier to compute!

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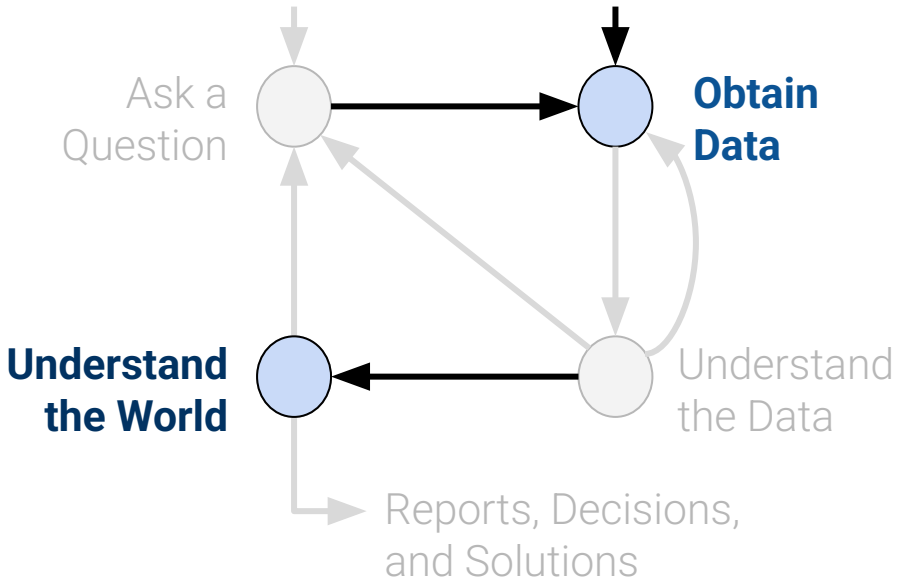
Summary

Understanding the sampling process is what lets us go from **describing the data** to **understanding the world**

Without knowing / assuming something about how the data were collected:

- There is no connection between the **sample** and the **population**
- The **data set** doesn't tell us about the **world behind the data**



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Participation Questions

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Q1

Statistical and Data Science methodologies can fix any type of bad sampling.

1. True
2. False

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Q1 Notes

- There are sophisticated ways to fix certain types of bad samples.
  - For example, if we undersampled a specific demographic, we can pay more attention to the people we *did* sample.
    - **Assuming** that we had a good sample within that demographic.

- Biased samples can (almost) never be salvaged.
  - We need a *very* trustworthy estimate of our bias.
  - If we can get such an estimate, why did we do a biased sample in the first place?
  - Garbage in, garbage out

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Q2

Random Sampling is the ***only*** way to get valid results.

1. True
2. False

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Q2 Notes

- There are very few "always" or "only" statements in this crazy world we live in.

- Common example: doing an SRS within each demographic of interest.
  - E.g., take an SRS of black people, white people, indigenous people, etc.
  - Not all sub-samples are possible
    - E.g. we'll never get a sample that only contains one group.
  - This can give *better* results than an SRS - we're guaranteed to have results for all people of interest.

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Q3

Which of the following is ***not*** an example of **statistical (sampling) bias**?

1. Sampling primarily from an elderly population to gauge interest in the Barbie movie
2. Sampling primarily from an elderly population to gauge interest in the Oppenheimer movie.
3. Interpreting the results differently due to preconceived notions and stereotypes.

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Q3 Notes

Bias in statistics refers to a **systematic difference** between your results and the truth.

- All samples will have random errors - your results won't be *exactly* the same.
- **Systematic difference**: your results would *tend* to be different if you repeated the sample.
  - E.g. usually overestimating the true value.

Again: samples will have error that we can quantify. Bias is when the estimate tends to overestimate (or underestimate).

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Q4

A larger sample will have less bias

1. This answer is invalid. Anyone who answers "A" was clearly not in class.
2. True
3. False

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Q4 Notes

Larger samples will *amplify* the bias!

- Larger samples have less variance in our estimates
  - More certain, but if there's bias then we're certain about the *wrong thing*.
- Always picture the Darth Vader toast!

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Q5

Which of the following statements is *false*?

1. Random sampling with and without replacement are very similar when the population is large.
2. All data sets must be accompanied with a description of the data collection strategy.
3. Low response rates on a survey are always a major cause for concern.

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